

**Math 110 – Homework 11 Answers**

# Math 110 – Fall 2026

## Homework 11

*Adjoints, self-adjoint and normal operators, and the spectral theorem*

**Suggested completion date: Friday, November 13**

Unless otherwise indicated, all vector spaces are over the field  $\mathbb{F}$ . Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

**Problem 1.** Let  $T : \mathbb{F}^n \rightarrow \mathbb{F}^n$  be defined by

$$T(z_1, \dots, z_n) = (0, z_1, \dots, z_{n-1}),$$

where  $\mathbb{F}$  is  $\mathbb{R}$  or  $\mathbb{C}$ . Find  $T^*$ .

**Problem 2.** Let  $T : V \rightarrow W$  be a linear map between finite-dimensional inner-product spaces. Prove that  $T$  is injective if and only if  $T^*$  is surjective.

**Problem 3.** Let  $T : \mathbb{F}^2 \rightarrow \mathbb{F}^2$  be the quarter-turn

$$T(x, y) = (-y, x).$$

- (a) Find  $T^*$ .
- (b) Prove that  $T$  is normal but not self-adjoint.
- (c) Find all eigenvalues of  $T$  when  $\mathbb{F} = \mathbb{R}$ , and when  $\mathbb{F} = \mathbb{C}$ .

**Problem 4.** Let  $P$  be a projector on a finite-dimensional inner-product space. Prove that the following are equivalent:

- (a)  $P$  is self-adjoint;
- (b)  $P$  is normal;
- (c)  $P$  is the orthogonal projection onto range  $P$ .

**Problem 5.** Let  $T$  be self-adjoint and let  $U$  be a  $T$ -invariant subspace. Prove that  $U^\perp$  is also  $T$ -invariant.

**Problem 6.** Consider

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

- (a) Find an orthonormal basis of  $\mathbb{R}^2$  consisting of eigenvectors of  $A$ .
- (b) Find an orthogonal matrix  $Q$  and diagonal matrix  $D$  such that

$$A = QDQ^T.$$

- (c) Use this diagonalization to find a formula for  $A^m$  for every positive integer  $m$ .

**Problem 7 (optional).** Let  $T$  be a normal operator on a finite-dimensional complex inner-product space and let  $p$  be a polynomial. Use the complex spectral theorem to prove that  $p(T)$  is normal.

## Solutions

### Problem 1

Solution notes

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### Problem 2

Solution notes

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### Problem 3

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### Problem 4

Solution notes

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**Problem 5**

**Solution notes**

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**Problem 6**

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**Problem 7**

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